

Nicholas Leger

2360 SW Archer Rd Apt 1003 | Gainesville, FL 32608 | 727-276-6355 | nicholasleger@ufl.edu

EDUCATION

University of Florida

Bachelor of Science in Mechanical Engineering
Minor in Engineering Innovation
Cumulative GPA: 3.47

2022 - December 2026

SKILLS

Software & Tools: SolidWorks (certified), Creo Parametric, Abaqus, Fusion 360, MATLAB, Arduino, LabView, MicroStation, SmartSketch
Engineering Skills: GD&T, Technical Drawing, FEA, Tooling Design, DFM, Prototyping, FMEA, Lean Manufacturing, Root Cause Analysis, Preventive Maintenance, Asset Optimization
Manufacturing & Automation: Manufacturing Process Planning, Manufacturing Execution Systems, CVML Cameras, Laser Systems

EXPERIENCE

Florida Department of Transportation, Gainesville, FL

March 2025-Present

Accelerated Pavement Testing Research Assistant

- Operated and maintained a Heavy Vehicle Simulator (HVS), implementing preventive maintenance strategies that reduced downtime and maximized testing efficiency
- Performed regular pavement deformation profiling using LabView, delivering data to support evaluation of pavement material
- Led the design and integration of a UV data collection system, enhancing simulation capabilities for real-world analysis

John Deere, East Moline, IL

May 2025-August 2025

Quality Engineering Intern

- Installed and programmed CVML cameras on the production line to automate quality checks and mitigate downtime
- Developed a 3D weld map to facilitate accurate quality-noting and improve quality conversation and resolution with operators
- Designed tooling and the manufacturing process to greatly improve the pulley alignment of the combine jackshaft pulley utilizing a new laser system saving over \$90,000 in direct warranty claims and improving customer satisfaction
- Covered quality issues for a zone of the main combine assembly line ensuring fast and efficient resolutions of quality issues
- Led quality communication at production meetings and discussed solutions with leadership to prevent future failures

John Deere, Augusta, GA

May 2024-August 2024

Manufacturing Engineering Intern

- Led factory fuel-fill reduction initiative, implementing process improvements that generated annual cost savings of over \$96,000.
- Designed and fabricated custom tooling to improve factory production efficiency and mitigate safety risks, applying DFM principles
- Engineered and prototyped a deluxe rear hoist system, achieving an estimated \$720,000 annual cost reduction from current offering
- Provided on-call production line support, applying root cause analysis to minimize downtime and maintain continuous operations
- Directed a standard time element audit, refining asset utilization and cost management strategies in alignment with Lean Manufacturing principles.

John Deere, Augusta, GA

August 2023-December 2023

Product Design Engineering Intern

- Designed, prototyped, and finalized drawings for 3 tractor add-on kits using GD&T principles expected to net over \$200,000/year
- Developed an improved factory creep-setting process, reducing post-maintenance costs by over 50%.
- Designed and collaborated with suppliers to expedite production of a hydraulic pipe spacer for production in the next MY

Savannah River Mission Completion – US DOE contract, Aiken, SC

May 2023-August 2023

Systems Engineering Intern

- Translated complex vendor diagrams into detailed 3D SolidWorks models, improving system integration and documentation.
- Updated and optimized Piping and Instrumentation Diagrams using MicroStation, ensuring compliance with engineering standards.
- Prepared and processed Design Change Forms in SmartSketch to implement engineering improvements in mechanical systems.

UF Solar Gators, Gainesville, FL

August 2022-May 2024

Aero Body Design Coordinator

- Led the fabrication of the solar car aero body, performing carbon fiber wet layups to minimize weight and air drag
- Designed and built safety-critical body and canopy latches, enhancing rider protection under varying failure conditions
- Conducted structural simulations in SolidWorks to analyze steel chassis performance under load, supporting data-driven design